

Building an Enterprise AI Workplace Assistant

Overview

Your team at **GlobalLogic** has been hired by **EnterpriseCorp**, a multinational organization with over **20,000 employees**, to build an AI-powered workplace assistant.

Instead of maintaining separate AI assistants for HR, IT, Finance, Travel, and Knowledge Management, EnterpriseCorp wants a **single intelligent platform** capable of understanding employee requests and routing them to the appropriate AI workflows.

The client expects a production-ready MVP within **two days**.

You will work as an AI consulting team, where each member is responsible for a different part of the solution.

Client Profile

Organization

EnterpriseCorp

- 20,000 Employees
 - Offices in 8 Countries
 - Hybrid Workforce
 - Multiple Business Units
 - Internal Helpdesk
 - Employee Portal
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Existing Technology

- Employee Directory
- HR Portal

- IT Helpdesk
- Leave Management System
- Travel Portal
- Knowledge Base
- Finance Reimbursement Portal
- Email & Teams

All systems expose mock REST APIs or JSON data for this capstone. It will be provided with the project

Current Business Challenges

Employees currently use multiple portals.

Examples:

- Apply Leave
- Reset Password
- Raise IT Ticket
- Search Company Policies
- Submit Expense Claim
- Book Business Travel
- Find HR Policies

Employees want **one AI assistant** instead of seven different systems.

Executive Goals

The CIO expects:

- Unified AI Experience
 - Intelligent Routing
 - Reduced Manual Support
 - Faster Employee Productivity
 - Modern AI Workplace
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Project Objective

Build an AI-powered Enterprise Assistant capable of understanding employee requests, deciding which department should handle them, executing the required workflow, and responding naturally.

Team Structure and role

Students should work in teams of **4 members**.

Team Member 1

AI Conversation Engineer

Responsible for:

- Chat UI
 - Prompt Engineering
 - Memory
 - Conversation History
 - System Prompt
 - User Experience
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Team Member 2

Tool & Integration Engineer

Responsible for building tools such as:

- Employee Lookup
 - Leave Request
 - Expense Claim
 - Travel Request
 - Password Reset
 - IT Ticket
 - Knowledge Search
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Team Member 3

Agent Engineer

Responsible for:

- LangGraph Workflow
 - Agent Logic
 - Tool Selection
 - Human-in-the-loop
 - Error Handling
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Team Member 4

Multi-Agent Engineer

Responsible for:

- Supervisor Agent
 - HR Agent
 - IT Agent
 - Finance Agent
 - Travel Agent
 - Knowledge Agent
-

All member

Evaluation & Deployment Engineer

Responsible for:

- LangSmith Tracing
 - Prompt Evaluation
 - Testing
 - Deployment
 - Documentation
-

Functional Requirements

The assistant should support requests like:

HR

- Apply Leave
 - Check Leave Balance
 - Holiday Calendar
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IT

- Reset Password
 - Raise Ticket
 - Check Ticket Status
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Finance

- Submit Expense
 - Check Reimbursement Status
-

Travel

- Request Business Travel
 - Estimate Budget
 - Generate Travel Plan
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Knowledge Base

- Company Policies
 - Security Guidelines
 - Employee Handbook
 - Remote Work Policy
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Level 1

AI Assistant

Build a conversational workplace assistant.

Features:

- Chat UI
 - Memory
 - Prompt Engineering
 - Guardrails
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Level 2

AI Agent

Convert the chatbot into an AI Agent capable of invoking tools.

Example tools:

`search_employee()`

`raise_ticket()`

`submit_leave()`

`submit_expense()`

`plan_travel()`

`search_policy()`

☐ The agent should automatically determine which tool(s) to invoke.

Level 3

Multi-Agent Architecture

Build a LangGraph-based workflow.

Recommended Agents:

Supervisor Agent

Receives every request.

Determines:

- HR
- IT
- Finance
- Travel
- Knowledge

Routes the request accordingly.

HR Agent

Handles:

- Leave
 - Holidays
 - Employee Information
-

IT Agent

Handles:

- Password
 - Laptop
 - Tickets
-

Finance Agent

Handles:

- Expenses
 - Reimbursements
-

Travel Agent

Handles:

- Trip Planning
 - Budget
 - Travel Requests
-

Knowledge Agent

Handles:

- Company Documentation
 - Internal Policies
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Human-in-the-loop

The system **must request confirmation** before:

- Leave Submission
 - Expense Submission
 - Ticket Creation
 - Travel Request
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LangSmith Requirements

Every team must:

- Enable tracing
- Record at least 10 conversations
- Analyze latency
- Review failed runs
- Improve one prompt based on trace insights

Deployment

Deploy the application using:

- Streamlit
 - Environment Variables
 - Production-ready README
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Deliverables

Each team must submit:

1. Working Application

Fully functional Streamlit application.

2. Source Code

Well-structured repository with proper folder organization.

3. Architecture Diagram

Illustrate:

- LLM
 - Tools
 - LangGraph Workflow
 - Multi-Agent Design
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4. LangSmith Report

Include:

- Traces
 - Prompt improvements
 - Evaluation observations
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5. Deployment

Live URL(use free cloud source like streamlit) (or local deployment demonstration if internet access is unavailable).

6. Technical Presentation

Include:

- Project Overview
- Setup Instructions
- Folder Structure
- Team Responsibilities
- Future Enhancements

PHASE 2 CLIENT ENHANCEMENT

EnterpriseAssist AI

Version 1 – Proactive AI Digital Employee

Background

Following the successful demonstration of the EnterpriseAssist AI MVP, EnterpriseCorp has approved the solution for pilot deployment across multiple business units. During the pilot, executives observed that employees primarily interacted with the assistant only after encountering a problem, such as raising an IT ticket, checking leave balances or submitting expenses.

The CIO has now initiated Phase 2 of the engagement with a strategic objective: transform EnterpriseAssist AI from a reactive virtual assistant into a Proactive AI Digital Employee capable of identifying potential issues before they impact employee productivity.

Business Context

Large enterprises generate thousands of daily events from HR, IT, Finance, Travel and Security systems. Employees often miss important deadlines because these events remain isolated inside departmental applications. The organization wants AI to continuously correlate enterprise events, understand employee context and recommend preventive actions instead of waiting for explicit user requests.

Phase 2 Client Requirement

Enhance EnterpriseAssist AI so that it continuously evaluates enterprise events, identifies situations requiring attention, prioritizes them according to business impact and presents employees with personalized recommendations. The assistant should explain why a recommendation is important and request approval before performing any business action.

Illustrative Enterprise Scenarios

- An employee's passport validity is insufficient for an approved international business trip.
- Mandatory cybersecurity awareness training expires tomorrow.
- A leave request overlaps with a planned production deployment.
- An expense claim is blocked because supporting invoices are missing.
- A laptop warranty expires next week while a critical project is in progress.
- A travel request exceeds the department's approved budget.
- Manager approval has been pending for more than five working days.

Business Objectives

- Reduce missed deadlines and manual follow-ups.
- Improve employee productivity through proactive guidance.
- Provide personalized recommendations using enterprise context.
- Reduce notification overload through intelligent prioritization.
- Increase adoption of EnterpriseAssist AI across business units.

Architecture Exploration

Teams are expected to design an event-driven GenAI architecture. The implementation approach is intentionally open. Students should justify how enterprise events are collected, correlated, prioritized and transformed into recommendations. Solutions may use rules, LLM

reasoning or hybrid approaches. Teams are encouraged to explain memory management, context handling, recommendation confidence and human approval mechanisms.

Design Challenges

- How should urgent and non-urgent events be distinguished?
- How should conflicting recommendations be resolved?
- How can unnecessary notifications be minimized?
- What enterprise data should influence personalization?
- How should the assistant explain the reasoning behind recommendations?
- How can recommendation quality be evaluated?

Expected Deliverables

1. Updated solution architecture highlighting proactive event processing.
2. Demonstration of at least five proactive recommendation scenarios.
3. Design justification describing prioritization and recommendation strategy.
4. Discussion of scalability, assumptions, limitations and future enhancements.

Evaluation Expectations

Assessment will focus on the quality of architectural thinking rather than the choice of framework. Teams should demonstrate clear reasoning, enterprise relevance, AI decision-making, user experience and the ability to justify design trade-offs. Multiple solution approaches are acceptable provided they address the business objectives.

Evaluation Criteria

Criteria	Weight(100)
Prompt Engineering & Conversation Design	15
LangChain & Tool Integration	20
Agent & LangGraph Workflow	20
Multi-Agent Collaboration	15
LangSmith Tracing & Evaluation	10
Deployment & Documentation	10
Team Collaboration & Final Presentation	10